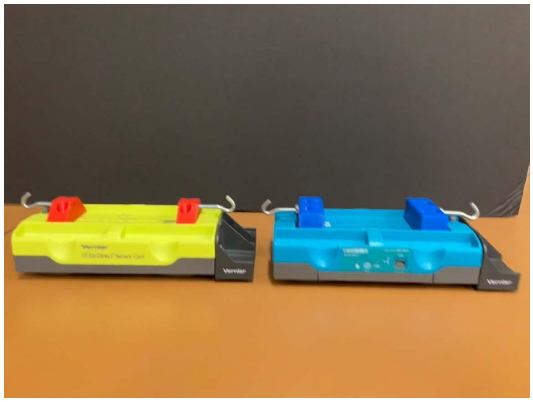
UMASS AMHERST · MIE 402 · FALL 2026 · EHSAN ROOHI

Lab 4 — Free-response analysis workflow

This is a supplementary explanation of the existing experiment, not a change in required hardware or seven-case procedure. Canvas and the released laboratory instructions remain authoritative for section-specific directions.

1. What to bring and record

Measure the total moving mass (cart plus all mounted and moving attachments), identify left and right calibrated springs, count actual brake **magnets**, record sample rate, release displacement and original CSV filename. A brake-assembly count is not interchangeable with a magnet count.



Actual Vernier sensor carts from the supplied MIE 402 course assets.

2. Calibrate each spring vertically

For hanging mass m_h [kg], static extension Δ_0 [m], spring stiffness k [N/m], and gravity g = 9.81 m/s², equilibrium requires $k\Delta_0 = m_h g$. For displacement q [m] from that equilibrium, $m_h\ddot{q} = m_h g - k(\Delta_0+q) = -kq$. Gravity cancels, leaving $m_h\ddot{q}+kq=0$. If Graphical Analysis fits force as $F(t) = a\sin(bt+c_{fit})+d$, its b is **angular frequency [rad/s]**, not frequency

in Hz, and c_{fit} [rad] is a phase—not the damping coefficient. For light damping and negligible spring mass, $k \approx m_h b^2$; otherwise investigate damping and effective mass.

3. Perform seven comparable cart runs

Run(s)	Configuration	Hypothesis to test
1	Base: original springs/mass, no added brake magnets	Measure residual damping; it is not zero
2–3	Add one mass, then a second; maintain same springs	Natural frequency decreases as mass increases
4–5	Two magnets, then four; maintain spring pair	Decay becomes faster; weigh all hardware
6–7	Return to base mass; use two alternate spring combinations	Natural frequency follows equivalent stiffness

Level the track; ensure both springs stay in pre-tension. In Vernier Graphical Analysis select the correct cart serial number, check the wheel encoder is rolling without slip, zero the position near equilibrium, start Collect and release *without a push*. Follow the supplied local sampling constraint (at most 25 Hz), collect multiple cycles and measurable decay, and export time and position to a distinct unmodified CSV for each run.

4. Derive the analysis quantities

Displacement $x(t)$ [m] is relative to horizontal equilibrium. Both preloaded springs oppose a small displacement: $\Delta F_1 = -k_1 x$ and $\Delta F_2 = -k_2 x$. Therefore $k_{eq} = k_1 + k_2$ [N/m]. Assuming a viscous-equivalent brake force $-c\dot{x}$, with c [N·s/m], Newton's law yields $m\ddot{x}+c\dot{x}+k_{eq}x=0.$

After dividing by m, define $\omega_n = \sqrt{(k_{eq}/m)}$ [rad/s] and $\zeta = c/(2m\omega_n)$ [dimensionless]. Roots of the characteristic equation are $-\zeta\omega_n \pm i\omega_d$ in the underdamped case, where $\omega_d = \omega_n\sqrt{(1-\zeta^2)}$ [rad/s]. Their real part gives $\alpha = \zeta\omega_n = c/(2m)$ [s^{-1}]; the imaginary part gives oscillation. The response is $x(t)=e^{-\alpha t}[C\cos(\omega_d t)+D\sin(\omega_d t)].$

5. Turn one CSV into measurements

1. Preserve raw CSV; select the free-response interval after hands have released the cart. Confirm position units (m, cm, or mm) and timestamps; do not apply a linear detrend blindly.
2. Subtract estimated equilibrium offset, and inspect for actual drift. Use reliable **same-sign** peaks above the noise floor.
3. Estimate damped period $T_d = (t_{last} - t_{first})/N$ [s], then $\omega_d = 2\pi/T_d$ [rad/s].
4. Fit $\ln(p/(1\ \text{m}))$ against peak time [s]. The slope is $-\alpha$ [1/s]. Alternatively use $\delta = \ln(p_i/p_{i+N})/N = \alpha T_d$.
5. Compute $\omega_n = \sqrt{(\omega_d^2 + \alpha^2)}$, $\zeta = \alpha/\omega_n$, and $c = 2m\alpha$ [N·s/m]. Compare predicted undamped frequency $\sqrt{(k_{eq}/m)}/(2\pi)$ [Hz].
6. Inspect time series, log-peak fit, full-response fit and its residuals, plus FFT magnitude versus frequency. The FFT peak near f_d is a check, not a standalone damping estimate.

6. MATLAB and evidence checklist

Run the main script `MIE402 Lab4 Analyze Free Response.m`. Never run internal helper functions separately. Read usage instructions. The current main script prompts for one CSV, hardware parameters, interval and position units, and saves figures and a summary beside the CSV. Its numerical results still require evaluation against actual measurements.

For the combined Labs 4–5 report, retain the original CSV and metadata for all seven configurations, labeled plots, fitted model and residual plot, calculated physical parameters and units, frequency differences, uncertainties and a short explanation of any mismatch. Cautions include encoder slip, spring slack, dry friction, irregular sample spacing, poor peak selection and variation in added hardware mass.

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